Chen Griner

Curriculum Vitae

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Education

2020 – 2024 **PhD in Communications Systems Engineering**, *Ben Gurion University of the Negev (BGU)*, Be'er Sheva, Israel.
Dissertation title: *Demand-Aware Topology Engineering for Reconfigurable Networks.* Award date: 22/08/2024

Supervisor : **Prof. Chen Avin**, *School of Electrical and Computer Engineering*, BGU (*Personal Web-page*).

2017 – 2020 **M.Sc. in Communications Systems Engineering**, *Ben Gurion University of the Negev (BGU)*, Be'er Sheva, Israel.
Thesis title: *The Complexity of Traffic Traces and its Application for Network Design.*

Supervisor : **Prof. Chen Avin**, *School of Electrical and Computer Engineering*, BGU.

2013 – 2017 **B.Sc. in Communications Systems Engineering**, *Ben Gurion University of the Negev (BGU)*, Be'er Sheva, Israel.
Comprehensive exposure to the core areas of computer and communication sciences, along with a final year project on Segment Routing in datacenters using SDNs.

Summary of Research & Academic Experience

Independent Project

2024 – 2025 ***Using Generative Pre-trained Transformers for Datacenter Research***.

- Conducted independent research on a generative-AI framework that synthesizes realistic **datacenter** network traffic using a modified GPT architecture.
- Published a short paper (poster) based on initial findings at IFIP Networking 2025.

PhD Researcher, BGU

2020 – 2024 ***Demand-Aware Topology Engineering for Reconfigurable Networks***.

- Published and presented 3 first-author papers in leading networking conferences and journals. One more paper is recently accepted and another under review.
- Explored different approaches (e.g., topology engineering) to optimize optical reconfigurable datacenter network (RDCNs), designs, and algorithms.
- Collaborated with researchers from Germany and the USA.

M.Sc Researcher, BGU

2017 – 2020 ***The Complexity of Traffic Traces and its Application for Network Design***.

- Developed an **information-theoretic** measure of structure in **network traffic**, called *"trace complexity"*. Trace complexity gives evidence to the extant possibilities for optimization in datacenter networks using dynamic networks.
- Published a single paper on ACM SIGMETRICS.

Teaching, BGU

2016 – 2023 **Teacher assistant with several courses related to computer networking**.

- Taught and led a practical *lab* course on computer networks and protocols
- Was a teacher assistant in courses on *Information Theory*, *Network Simulation*, and *Distributed Algorithms*.

Full Publications List

The following includes my publications and preprints. Note: *download links* lead to my personal website, where all my publications are available.

Journal Articles

2025 **Chen Griner**, and Chen Avin, "Integrated Topology and Traffic Engineering for Reconfigurable Datacenter Networks", *Computer Networks*, page 111803, Elsevier. **(Impact Factor: 4.4)**.

- **DOI & Download link:***doi.org/10.1016/j.comnet.2025.111803*

2024 **Chen Griner**, Chen Avin, and Gil Einziger. "Beyond matchings: Dynamic multi-hop topology for demand-aware datacenters", *Computer Networks*, page 110143, Elsevier. **(Impact Factor: 4.4)**.

- **DOI & Download link:** *doi.org/10.1016/j.comnet.2023.110143*

2022 **Chen Griner**, Stefan Schmid, and Chen Avin. "Cachenet: Leveraging the principle of locality in reconfigurable network design", *Computer Networks*, , volume 204, page 108648. Elsevier. **(Impact Factor: 4.4)**.

- **DOI & Download link:** *doi.org/10.1016/j.comnet.2021.108648*

In Conference Proceedings

2025 **Chen Griner**, "Effectively mimicking datacenter traffic patterns using transformers", *IFIP Networking Conference*, pages 1–3, IEEE. **(Ranking: A)**.

- **Note:** Short paper.
- **DOI & Download link:** Proceedings not yet published: *Personal-website*

2021 **Chen Griner***, Johannes Zerwas*, Andreas Blenk, Manya Ghobadi, Stefan Schmid, and Chen Avin, "Cerberus: The power of choices in datacenter topology design-a throughput perspective", *Proc. ACM Meas. Anal. Comput. Syst.*, Volume 5, pages 1–33, ACM New York, NY, USA. **(Ranking: A*)**.

- **Note:** authors marked with * contributed equally to the completion of this work. First published at ACM SIGMETRICS conference.
- **DOI & Download link:** *doi.org/10.1145/3491050*

2021 **Chen Griner** and Chen Avin, "Cachenet: Leveraging the principle of locality in reconfigurable network design", *IFIP Networking Conference* , pages 1–3, IEEE. **(Ranking: A)**.

- **Note:** Short version of the later published Journal version.

2020 Chen Avin, Manya Ghobadi, **Chen Griner**, and Stefan Schmid, "On the complexity of traffic traces and implications", *Proc. ACM Meas. Anal. Comput. Syst.*, Volume 4, pages 1–29, ACM New York, NY, USA. **(Ranking: A*)**.

Note: Name order is alphabetical. First published at the ACM SIGMETRICS conference.

DOI & Download link: *doi.org/10.1145/3379486*

Preprints

2024 **Chen Griner**, "Harnessing generative pre-trained transformer for datacenter packet trace", *arXiv preprint arXiv:2501.12033*.

Description: Full and technical version of the IFIP short paper.

DOI & Download link: *doi.org/10.48550/arXiv.2501.12033*

2024 Johannes Zerwas*, **Chen Griner***, Stefan Schmid, Chen Avin, "D3: An Adaptive Reconfigurable Datacenter Network", *arXiv preprint arXiv:2406.13380*.

Notes: Under Submission & authors marked with * contributed equally to the completion of this work.

DOI & Download link: *doi.org/10.48550/arXiv.2406.13380*

Scholarships & Awards

2020 – 2023 **BGU**, Received *Negev* fellowship for outstanding doctoral students.

2020 **BGU**, Received an award on behalf of the Dean of the Faculty of Engineering Sciences.

2020 **BGU**, Graduated with highest honors from M.Sc (summa cum laude).

2019 **BGU**, Received Certificate of Excellence for first year of M.Sc.

2017 **BGU**, Received Certificate of Excellence for last year of B.Sc.

Presentations at Universities & Conferences

May 28th 2025 **Poster**, *Effectively Mimicking Datacenter Traffic Patterns Using Transformers*, IFIP Networking Conference 2025 , Limassol, Cyprus..

June 7th 2022 **Paper**, *Cerberus: The Power of Choices in Datacenter Topology Design - A Throughput Perspective*, ACM SIGMETRICS 2022 , Mumbai, India & Online.
(*Video presentation*)

June 24th 2021 **Poster**, *CacheNet: Leveraging the Principle of Locality in Reconfigurable Network Design*, IFIP Networking Conference 2021 , Online.

June 8th 2020 **Paper**, *On the Complexity of Traffic Traces and Implications*, ACM SIGMETRICS 2020 , Boston, Massachusetts, USA & Online.
(*Video presentation*)

July 5th 2019 **Poster**, *The Complexity of Traffic Traces*, DMBI conference , BGU,Be'er Sheva, Israel..

April 16th 2019 **Poster**, *The Complexity of Traffic Traces*, New England Network and Systems Day , MIT, Boston, Massachusetts, USA.

Teaching

Fall, 2022 – Fall, 2021 **Head Lab Instructor** | **Computer Networks Design Laboratory**, CSE BGU.
Subjects: Running network and protocol experiments on an emulated network of Cisco routers. Including DNS, DHCP, VLAN, MPLS, IPv6, BGP, PIM, IGMP, and RTP. Developing a client-server application using C socket programming running on a Linux VM.

Spring, 2022 – Spring, 2021 **Head Lab Instructor** | **Computer Networks Laboratory**, CSE BGU.
Subjects: Running network and protocol experiments on an emulated network of Cisco routers. This included the configuration of IP routers in IP networks. Routing Algorithms, OSPF, RIP. Transport layer protocols- TCP/UDP. Programming of simple client-server applications with C socket programming running on a Linux VM.

Fall, 2020 – Fall 2017 **Head Lab Instructor** | **Computer Networks 2 Laboratory**, CSE BGU.
Subjects: Running network and protocol experiments on an emulated network of Cisco routers. Including OSPF, RIP, BGP, PIM, and TCP/UDP. Developing a client-server application using C socket programming running on a Linux VM.

Spring, 2020 **Lab Instructor** | **Simulation in Networks**, CSE BGU.
Modeling and analysis in network research. Acquiring principles of simulation, design, writing, and building simulators, and the use of simulation packages from Omnet++ for modeling and analysis.

Spring, 2020 **Teaching assistant** | **Introduction to Information Theory**, CSE BGU.
Introductory Subjects to Information Theory, including: Information transfer rate, entropy and mutual information, lossless coding, noisy channels, coding theorems for discrete memoryless channels, channel capacity, data processing theorem, error bounds and source coding.

Spring, 2019 **Teaching assistant** | **Distributed algorithms**, CSE BGU.
Subjects: principles of distributed computing. Various failure models. Network search, leader election in a ring and arbitrary network, network routing, transaction management, and others.

Spring, 2018 **Teaching assistant** | **Introduction to Information Theory**, CSE BGU.
Subjects: As mentioned before.

Reviewing activity

2025	IEEE/ACM Transactions on Networking.
2024	IEEE Transactions on Network and Service Management.

Technical skills & Languages

Programming:	Python (including PyTorch), C, Wolfram Mathematica.
Networking:	OMNeT++, GNS3, Wireshark.
Languages:	English, Hebrew.